GENERALIZED ESTIMATION SYSTEM

Version 2.03

Methodology Guide – Sampling variance with Taylor linearization

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G-Est

Statistics Canada Statistique Canada

Canada

Contents

1 Introduction

The generalized estimation system, G-Est is a collection of SAS macros that perform various weighting, point estimation, and variance estimation tasks. In version 1 of G-Est, modules are available to perform calibration using the calibration module, Gest_Calibration; point estimation using the estimation module, Gest_PointEstimation; and variance estimation using the variance module, Gest_Variance. The types of variances that G-Est can estimate are not limited to the sampling variance, but include variance due to nonresponse and imputation. This latter functionality is available by integrating the SAS application SEVANI (Beaumont et al., 2010; Beaumont and Bissonnette, 2011). Further information on the methodology behind the calibration module and the variance due to nonresponse and imputation functionality of the variance module are available in the methodology guides for those modules (Generalized Systems Methods, 2014a; Generalized Systems Methods, 2014b).

This document describes the methodology behind the estimation module, Gest_PointEstimation and the sampling variance component of the variance module, Gest_Variance. The Gest_PointEstimation module computes the point estimates of the domain size, total, mean, and ratio. The Gest_Variance module computes the sampling variance estimates of the parameters. The computations behind estimation are discussed in conjunction with sampling variance since the concepts are closely related and some formulas overlap.

The methods used in G-Est, and described in this document, for estimating parameters and their sampling variance are classical methods for which more detailed development can be found in the survey sampling literature. The relevant references include Särndal et al. (1987), Deville and Särndal (1992), Estevao et al. (1995), Estevao and Särndal (2006), Estevao and Särndal (2009), and Generalized Systems Methods (2011). As well, many of the methods have previously been implemented in the systems GES and StatMx, the predecesors to G-Est at Statistics Canada, so the official documentation for those systems is relevant.

In the next section on Parameter Estimation, the formulas for different estimation scenarios are presented. In the same section, some background information is provided on the inputs necessary to provide the sampling information to the modules. In particular, a discussion is provided on the domain and parameter files, as well as the input parameters, OutOfScopeExpression and CountLevelID. In Section 3, on variance estimation, details are provided on the linearization of the parameters for which the sampling variance is to be computed and the computations of the residuals under one-phase and two-phase designs. The variance formulas for the specific sampling design available in G-Est are also presented in the section.

2 Parameter Estimation

G-Est computes domain estimates of the following population parameters: totals, means, sizes, and ratios. Point and variance estimation are available when the variables of interest are numerical variables. If the variables of interest are categorical variables, only point estimation is available. If the estimation unit does not coincide with the sampling unit, estimation of the size or mean can occur at either the estimation or sampling unit *count level*. In G-Est, parameter estimation is carried out by the module Gest_PointEstimation using files containing the domain and parameter definition information.

In order to compute domain estimates, domain and parameter files must be provided to G-Est. These files provide the information necessary to compute the parameter estimates and their corresponding variances. The same file can be used in both Gest_PointEstimation and Gest_Variance. Given a domain file and a parameter file, G-Est forms the Cartesian product of the domains and the parameters and estimates are then computed for the resulting combinations.

Example 1

Suppose the domain file contains one line for province and another line for industry. Further suppose the parameter file contains one line for the total revenue and another line for mean revenue. G-Est will then compute estimates of total revenue and mean revenue for each level of province and each level of industry.

2.1 Domain Files

Domains are provided to G-Est through the use of a domain file. The domain file contains records with each record defining a domain group or domain family. These are specified in the variable DomainGroup. A domain group can be modified through the use of a domain expression, which is specified by a valid SAS expression in the variable DomainExpression. The expression is evaluated within each domain group. The domain considered by G-Est is then the conjunction of DomainExpression and each level of DomainGroup.

The variable DomainGroup contains one or more variable names corresponding to the domain variables described in the sample file section. Specifying the domain group in this way, as a series of variables, would imply that the user wanted all combinations of settings of the indicated variables. If a variable that is part of the definition of a domain group is denoted by d_i, for $i = 1, \ldots, D$ and if d_i has J_i settings, then the total number of domains would be $J = J_1 \times J_2 \times \cdots \times J_D$. Note, however, that some of these domains may be empty, either because there were no population elements in the domain or none of the elements in the domain were sampled. Therefore, the actual number of domains that would need to be computed could be lower than J.

The domain expression permits the user to indicate additional precision in the domain definition by providing a SAS expression that would return all the units that satisfy specific criteria. This expression would be evaluated within the corresponding domain groups. Subsets of the domain groups would be derived in this way. Valid SAS expressions would involve variables available on the sample file and settings that are eligible for the indicated variables.

Example 2

Suppose that, on a sample file of enterprises, we have the variables province, industry, and size, where the industry variable contains seven classifications and the size variable contains three levels corresponding to large, medium, and small. We could then specify the following domains.

DomainID	DomainGroup	DomainExpression	DomainDescription
1	province industry size		All domains
2	province industry size	sales > 100	
3	province industry	sales > 100 and size = "medium"	Large sales from medium companies
4	province	industry = "NAICS415" and size = "large"	

Since there are ten provinces, seven industries, and three sizes, the first row of the domain definitions file indicates that there will be up to 210 domains that will contain parameter estimates. The second row adds to these domains by specifying that, for each of the 210 combinations of province, industry, and size variables, only enterprises with sales exceeding 100 million dollars will be included in the calculation. In the third domain specification, only the combinations of province and industry are considered, along with the further specification of sales exceeding 100 million and enterprise size being medium. This specification happens to be a subset of the second domain definition. Finally, the fourth line precisely defines the domain as a specific industry and size in each province. In total, from the four rows of the foregoing domain definitions file, we can expect up to 500 domains to be output for each parameter estimate requested.

2.2 Specifying Out of Scope Units

In some situations, it is desired that certain units or groups of units are excluded from the computation of the estimates because these units are considered out of scope. However, although these units may be safely ignored from a methodological point of view for parameter estimation, they may still need to be taken into account in the estimation of the variance. Therefore, units that are out of scope must remain on the sample file. The `OutOfScopeExpression` parameter in `Gest_PointEstimation` and `Gest_Variance` further modifies the domain group and domain expression by requiring certain units to be excluded from estimation, but to retain these units on the file so that they contribute to the computation of the sampling variance. The out of scope expression is global in nature and is applied to all units in the sample for each DomainID.

Example 3

Consider the domain file described in Example 2. Suppose, in addition to the domains defined there, we wish to exclude from estimation units where the out of scope variable, `OOS`, is equal to 1. This is accomplished by specifying the expression `OOS = 1` in the `OutOfScopeExpression` as follows:

```
%Gest_PointEstimation(
    ...,
    OutOfScopeExpression = OOS = 1,
    ...
);
```

Since `OutOfScopeExpression` is a global parameter, if there are multiple domain groups defined in the domain file, the out of scope expression applies to all of them.

2.3 Parameter Files

The parameters and the variables of interest are specified in the parameters file. Each record contains a value for the variable `ParameterDefinition` that informs `G-Est` what parameter has been requested. The available parameters that can be provided to `G-Est` are as follows.

Parameter	**ParameterDefinition**
Size	Size
Total of *variable*	total(*variable*)
Mean of *variable*	mean(*variable*)
Ratio of *variable1* and *variable2*	ratio(*variable1*, *variable2*)

The parameter file also contains a variable that optionally defines whether a given variable of interest is a categorical variable. If it is, then the parameter definition will apply to the count of each level of the variable.

Example 4

Suppose we wish to estimate the domain sizes and the domain totals, means, and ratios of several variables. If the sample file contains the values of the variables: sales, expenses, salary, and type, where type is a categorical variable indicating one of three types of firm; we can specify the following parameter definitions file.

ParameterID	ParameterDefinition	CategoricalVariables	ParameterDescription
1	size		Domain size
2	total(sales)		Total sales in current year
3	mean(salary)		Average salary
4	ratio(sales, expenses)		Ratio of sales to expenses
5	total(type)	type	Size of each type
6	mean(type)	type	Proportion of each type

Then, for each of the domains specified in the domain definitions file, ten parameter estimates will be computed. The first one, size, would result in the estimated size of each domain. The next estimate is the total to be estimated for the current value of sales. The third estimate is for the average salary. The fourth estimate computes a ratio of sales to expenses. The fifth record requests that `G-Est` computes the estimated size of the population that is made up of each type. Since there are three types, three estimates will be produced. Finally, the last record requests that the estimated proportion of each type will be produced. Again, since there are three types, three such estimates will be produced.

2.4 Specifying the Count Level

It may be the case that the sampling unit differs from the estimation unit, as may occur in a single-stage clustered design. In `G-Est` the sample file can be specified at either the sampling unit level or the estimation unit level. If the sample is specified at the estimation unit level and this level is different from the sampling unit level, the user may choose to compute the domain size and mean at either the sampling or estimation unit level. This is indicated to `G-Est` through the `CountLevelID` parameter. If `CountLevelID` is specified to be the sampling unit ID, then for the domain size and mean, the cardinality of the units used in the computations will be based on the sampling units. If `CountLevelID` is specified to be the estimation unit or is omitted, `G-Est` will compute the domain size and mean at the level of file. Note that this implies that if the sample file is not at the level of the sampling unit, the level of the file is assumed to be at the level of the estimation unit.

While it is possible to specify a count level that is intermediate to the sampling unit and the estimation unit in `Gest_PointEstimation`, this is not supported in `Gest_Variance`. In the foregoing, we mention the size and mean explicitly as the domain total and ratio are not affected by the count level when they are computed over numeric variables.

Example 5

Consider the parameter file in Example 4. Suppose the sample file was at the location level, but the sample was drawn at the enterprise level. In addition, the domains are at the location level and we wish to have the domain sizes and means computed at that level. If the parameter `CountLevelID` is left blank, by default, `G-Est` will perform the calculations as desired. Alternatively, to explicitly request that the location be used as the count level, we can call the estimation module as follows:

```
%Gest_PointEstimation(
     ...,
     CountLevelID = location_id,
     ...
);
```

2.5 Parameter Estimation When the Sampling Unit Equals the Estimation Unit

In this section, formulas for the available `G-Est` parameters are given. Because `G-Est` permits the input sample file to be at either the sampling unit level or the estimation unit level, computations for the domain size, domain mean, and those involving categorical variables will be different depending on the structure of the sample file and the specified count level. In this section, we assume that the sampling unit and the estimation unit coincide. The case where they differ is treated in Section 2.6. Similarly, the estimation of parameters based on categorical variables, which is supported only in the estimation module, is discussed in Section 2.7.

Let the population of interest be denoted by U and the sample be denoted by s. If the population is comprised of N elements $1, \dots, k, \dots, N$, then the sample is formed by the elements $k \in s$. Now let possibly overlapping subpopulations of U be formed. We indicate the d-th subpopulation as U_d and refer to it as the d-th domain. Define the domain indicator function I_{dk} as

$$I_{dk} = \begin{cases} 1, & \text{if } k \in U_d \\ 0, & \text{if } k \in U - U_d \end{cases}. \qquad (2.1)$$

Let y_k be the variable of interest for unit k. Define the extended domain variable y_{dk} as

$$y_{dk} = \begin{cases} y_k, & \text{if } k \in U_d \\ 0, & \text{if } k \in U - U_d \end{cases}. \qquad (2.2)$$

Suppose a given combination of DomainGroup and DomainExpression results in the domains U_d. Let the set of units defined by OutOfScopeExpression be given by U_{oos}. Then, the effective domains for which estimates will be computed are given by $U_\delta = U_d \cap U_{oos}^c$. Consequently, the domain indicator and extended domain variables defined by Equation (2.1) and Equation (2.2) will now refer to the new domains. Let the new indicator be denoted by $I_{\delta k}$ and equal to 1 if k is a member of U_δ and 0 otherwise. Similarly, $y_{\delta k}$ equals y_k if k is a member of U_δ and 0 otherwise. Since the OutOfScopeExpression is global and applies to all units in the requested domains, if the domains U_d are also indicated in the domain file, then estimates will also be computed for $U_{\delta'} = U_{d'} \cap U_{oos}^c$.

Let the estimation weights be given by w_k. The domain parameters and their estimates are then given by the following equations.

Size

The true size in domain U_d is then given by

$$N_d = \sum_{k \in U} I_{dk} \,.$$

The estimate of the domain size is

$$\hat{N}_d = \sum_{k \in s} w_k I_{dk} \,.$$

Total

The true total of the variable y in domain U_d is given by

$$Y_d = \sum_{k \in U} y_{dk} \,.$$

The estimate of the domain total of y is

$$\hat{Y}_d = \sum_{k \in s} w_k y_{dk} \,.$$

Mean

The true mean of the variable y in a domain U_d is given by

$$\bar{Y}_d = \frac{\sum_{k \in U} y_{dk}}{N_d} \,.$$

The estimate of the domain mean of y is

$$\hat{\bar{Y}}_d = \frac{\sum_{k\in s} w_k y_{dk}}{\hat{N}_d}.$$

Ratio

The true ratio R_d of two variables Y_1 and Y_2 in domain U_d is given by

$$R_d = \frac{Y_{1d}}{Y_{2d}} = \frac{\sum_{k\in U} y_{1dk}}{\sum_{k\in U} y_{2dk}}.$$

The estimate of the domain ratio of y_1 and y_2 is

$$\hat{R}_d = \frac{\hat{Y}_{1d}}{\hat{Y}_{2d}} = \frac{\sum_{k\in s} w_k y_{1dk}}{\sum_{k\in s} w_k y_{2dk}}.$$

2.6 Parameter Estimation When the Sampling Unit Differs from the Estimation Unit

This section provides some formulas and examples to illustrate the calculation of the various parameters available in `G-Est` when the sample file is given at the estimation unit level, but sampling was done at a higher level. The count level is set at the either the sampling (cluster) or estimation (element) level. By count level, we mean the unit used to derive the cardinality of the set for which estimates will be computed. Thus, for example, a count level equal to the estimation unit implies that the domain mean is computed as the domain total per estimation unit.

Note that if the file is given at the sampling unit level, then the estimation unit is assumed to equal the sampling unit. In this case, there is no need to specify the count level and the formulas used for parameter estimation are covered by Section 2.5. However, if the file is at the estimation unit level, then the count level can be either at the sampling unit or the estimation unit level.

Denote the sampling units by k and the estimation units by j, where $j \in k$. If the sample file is at the estimation unit level, we assume that the domain variables are also at the estimation unit level. Let d be a level of a domain U_d. The domain indicator for an estimation unit j in domain U_d is then given by I_{dj}, where I_{dj} equals 1 in the domain and 0 otherwise.

Size

The count level is the estimation unit

The estimated size of the domain U_d at the sampling level when the count level is the estimation level is given by

$$\hat{N}_d = \sum_{k\in s} w_k \sum_{j\in k} I_{dj} = \sum_{j\in s} w_j I_{dj},$$

where the j are the estimation units and the k are sampling units. Although the notation w_j refers to the sampling weights at the estimation level, the weights must be equal for all estimation units j in a given sampling unit k.

The count level is the Sample Unit

If the file is given at the estimation level, but the count level is set to the sampling unit, the size is computed as follows. Let the sampling unit k contain the j estimation units. If there is at least one estimation unit j in sampling unit k that is in the domain U_d, we include that sampling unit in the estimate of the domain size. That is, we "count" the sampling unit k if $\max_{j\in k} I_{dj} = 1$. The estimated size of the domain U_d at the sampling level when the sample file is at the estimation level is then given by

$$\hat{N}_d = \sum_{k\in s} w_k \max_{j\in k} I_{dj}.$$

Mean

The count level is the estimation unit

The estimated domain mean of a numeric variable is computed as

$$\hat{\bar{Y}}_d = \frac{\sum_{k\in s} w_k \sum_{j\in k} y_{dj}}{\sum_{j\in s} w_k \sum_{j\in k} I_{dj}} = \frac{\sum_{k\in s} \sum_{j\in s} w_j y_{dj}}{\sum_{k\in s} \sum_{j\in k} w_j I_{dj}},$$

where $w_j = w_k$ for all $j \in k$.

The count level is the sampling unit

The estimated domain mean of numeric variable is computed as

$$\hat{\bar{Y}}_d = \frac{\sum_{k\in s} w_k \sum_{j\in k} y_{dj}}{\sum_{j\in s} w_k \max_{j\in k} I_{dj}}$$

2.7 Parameter Estimation of Categorical Variables

If the variable of interest y is categorical, let $U_{(dc)}$ be the domain defined by the intersection of the domain U_d and the category c. Then, the indicator variable $I_{(dc)j}$ equals 1 if the unit j is a member of $U_{(dc)}$ and 0 otherwise.

Total

In the case where the variable of interest y is a numeric variable, the total Y_d is affected neither by the structure of the sample file, nor by the choice of count level. In all cases, any estimate of the total will yield the same result. If, however, y is a categorical variable, then computing the total of the variable is equivalent to computing the size of each category of the variable. Therefore, the file structure and count level impact on the calculation as in the case of the domain size.

If the file is at the estimation unit level and the count level is the estimation unit, then the total $\hat{Y}_{(dc)}$ of a categorical variable for c is given by

$$\hat{Y}_{(dc)} = \sum_{k\in s} w_k \sum_{j\in k} I_{(dc)j} = \sum_{j\in s} w_j I_{(dc)j},$$

where $w_j = w_k$ for all $j \in k$.

If the file is at the estimation unit level and the count level is the sampling unit, then the total $\hat{Y}_{(dc)}$ of a categorical variable for c is given by

$$\hat{Y}_{(dc)} = \sum_{k\in s} w_k \max_{j\in k} I_{(dc)j}.$$

Mean

If the file is at the estimation unit level and the count level is the estimation unit, the estimated mean of a categorical variable for category c is

$$\hat{\bar{Y}}_{(dc)} = \frac{\sum_{k\in s} w_k \sum_{j\in k} I_{(dc)j}}{\sum_{k\in s} w_k \sum_{j\in k} I_{dj}} = \frac{\sum_{j\in s} w_j I_{(dc)j}}{\sum_{j\in s} w_j I_{dj}}.$$

If the file is at the estimation unit level and the count level is the sampling unit, the estimated mean of a categorical variable for category c is computed as

$$\hat{\bar{Y}}_{(dc)} = \frac{\sum_{k\in s} w_k \max_{j\in k} I_{(dc)j}}{\sum_{k\in s} w_k \max_{j\in k} I_{dj}}.$$

3 Variance Estimation

In this section, we present the methodology used by `G-Est` to compute the estimates of sampling variance. The approach taken in version 1 of `G-Est` is to use Taylor linearization. Using linearization, the non-linear calibration estimator is approximated by a linear function and the variance is then taken of the linear function. As mentioned above, there are four parameters for which we require variance estimates: size, total, mean, and ratio. Since the domain mean (with estimated domain size) and ratio are also non-linear, linearization is also required there. So, if design weights were used to estimate these parameters, the standard variance estimators (e.g., the Horvitz-Thompson variance estimator) could be used, possibly requiring linearization in the case of the mean and ratio. If calibration weights were used, we require linearizing again to take into account the calibration weights. Section 3.4 explains the linearization of the parameters. Here, the linearization for the calibration weights is briefly described.

The calibration weights used in `G-Est` are based on minimizing the chi-squared distance. That is, the following minimization problem is solved.

Minimize

$$\sum_{k\in s} G_k(w_k, a_k) = \sum_{k\in s} \frac{c_k(w_k - a_k)^2}{a_k}, \text{ with respect to } w_k$$

Subject to

$$\sum_{k\in s} w_k \mathbf{x}_k I_{pk} = \mathbf{t}_p, \qquad p = 1, \ldots, P.$$

Here, a_k are the design weights, c_k is the regression adjustment weight, $\mathbf{x}_k$ is the vector of auxiliary variables, I_{pk} is the calibration group membership indicator for group p, and $\mathbf{t}_p$ is the totals for group p.

The calibration weights formed in this way are also known as generalized regression (GREG) weights. Consider a one-phase design. When no boundary constraints are imposed, the GREG weights can be written in closed form, as follows.

$$w_k = a_k \left[1 + (\mathbf{X} - \widehat{\mathbf{X}})' \left(\sum_{k\in s} a_k \mathbf{x}_k \mathbf{x}_k' / c_k \right)^{-1} \mathbf{x}_k \right],$$

where $\mathbf{X} = \sum_{k\in U} \mathbf{x}_k$ and $\widehat{\mathbf{X}} = \sum_{k\in s} a_k \mathbf{x}_k$. Let u_{dk} be the linearized value for domain d for the parameter we want to estimate, then the calibration estimator can be written as

$$\hat{Y}_w = \sum_{k\in s} w_k u_{dk} = \hat{Y}_a + (\mathbf{X} - \widehat{\mathbf{X}})' \widehat{\mathbf{B}}, \tag{3.1}$$

where $\widehat{\mathbf{B}} = (\sum_{k\in s} \mathbf{x}_k \mathbf{x}_k')^{-1} \sum_{k\in s} a_k \mathbf{x}_k u_{dk}$. Let $\mathbf{B}$ be the true value of $\widehat{\mathbf{B}}$. Adding and subtracting $\mathbf{B}$ to Equation (3.2) and rearranging terms, we obtain

$$\hat{Y}_w = \mathbf{X}'\mathbf{B} + (\hat{Y}_a - \widehat{\mathbf{X}}'\mathbf{B}) - (\widehat{\mathbf{X}} - \mathbf{X})(\widehat{\mathbf{B}} - \mathbf{B}).$$

Denote the individual terms of $\left(\hat{Y}_a - \hat{\mathbf{X}}'\mathbf{B}\right)$ by $e_{dk} = u_{dk} - \mathbf{x}_k'\mathbf{B}$**.** The lower order term $\left(\hat{\mathbf{X}} - \mathbf{X}\right)\left(\hat{\mathbf{B}} - \mathbf{B}\right)$ is then dropped (c.f. Estevao and Sarndal, 2006, p131). And since $\mathbf{X}'\mathbf{B}$ is a constant, its variance is zero, so the variance of $\hat{Y}_w$ reduces to $\mathrm{Var}(e_k)$. The exact form and estimator of the variance also need to take into account the sample design. This is described in more detail in Sections 3.6 and 3.7 .

3.1 Calibration Group Files

If calibration weights were used to do the parameter estimation, it is necessary to supply `G-Est` with the calibration group file used in the calibration module. The calibration file is used by `G-Est` to create the auxiliary data matrix, which in turn is used to compute the residuals for variance estimation.

The calibration group file contains information on both the calibration groups and the auxiliary variables. In `G-Est`, the calibration groups may be overlapping and not necessarily exhaustive. This is the most general case where there are multiple sets of calibration groups and at least one set is not exhaustive. The calibration groups are not mutually exclusive, so a unit may belong to more than one group. It may also be the case that not all the calibration groups in a set of groups that partition the sample were used in calibration. Or it may also be the case that, even though all the groups in a set were used, not all of the groups used the same auxiliary variables.

Example 6

Calibration groups are formed separately by combinations of province and industry and auxiliary variables are revenues and expenses. The following calibration group file was used in `Gest_Calibration`.

CalibrationGroupID	CalibrationGroup	CalibrationGroupExpression	AuxiliaryVariables	…
1	province industry		revenue	…
2	province industry	province ne ‘PE’	expenses	…

If there are five industries, the first line results in 15 calibration groups using revenue as the auxiliary variable. In the second line, the `CalibrationGroupExpression` is used. The `CalibrationGroupExpression` specifies units or groups of units in the sample file for which calibration is to be done. Units not meeting the criteria in the `CalibrationGroupExpression` will not have their weights calibrated. Thus, for each province and industry combination, only units for which the province is not equal to `PE' are part of the calibration.

If a two-phase design is used and two-phase calibration was done, two sets of calibration group files would be needed. See Section 3.7.

3.2 Specifying the CalibrationExclusionExpression

The `CalibrationGroupExpression` specifies units or groups of units in the sample file for which calibration is to be done. Units not meeting the criteria in the `CalibrationGroupExpression` will not have their weights calibrated. By contrast, the `CalibrationExclusionExpression` specifies the units or groups of units for which calibration is not to be done. Units not meeting the criteria in the `CalibrationExclusionExpression` will have their weights calibrated.

The `CalibrationExclusionExpression` and `CalibrationGroupExpression` are not directly interchangeable in usage without modification of the auxiliary totals. When `CalibrationGroupExpression` is used, it is assumed that the auxiliary totals in the `TotalsFile` pertain only to the combination of `CalibrationGroup` and `CalibrationGroupExpression`. When `CalibrationExclusionExpression` is used, `G-Est` automatically adjusts the totals internally by subtracting the sum of the initial weights of the excluded units from the totals specified in the `CalibrationGroupFile`. However, as the sampling variance module does not require the totals to estimate the sampling variance, we can ignore this. This implies that when `G-Est` automatically adjusts the totals to accommodate the `CalibrationExclusionExpression`, we treat the resulting totals used in calibration as known.

Example 7

Suppose the sample file contains *take-all* units for which we do not want to do calibration. This is indicated to the variance module as follows.

```
%Gest_Variance(
     ...,
     CalibrationExclusionExpression = takeall = 1,
     ...
);
```

3.3 The Auxiliary Data Matrix

In the more general case where the calibration groups are not mutually exclusive or exhaustive, we must form the auxiliary data matrix $\mathbf{X}$, where $\mathbf{X}$ is the horizontal concatenation of the $\mathbf{x}_k$. Let $\mathbf{x}_k = (x_{k1}, \dots, x_{kQ})'$ be the vector of Q auxiliary variables for element k. We have

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1' \\ \vdots \\ \mathbf{x}_n' \end{pmatrix}. \tag{3.2}$$

The auxiliary data vector for element k will now be defined in relation to the element's membership in the calibration groups. If the auxiliary variables are represented again by $x_{k1}, \dots, x_{kQ}$, where the x_{kq} are numeric, continuous, and available on the input sample file, then each x_{kq} will be multiplied by the calibration group membership indicator for $p = 1, \dots, P$. In addition, it may be the case that calibration to counts was done in some calibration groups. The vector $\mathbf{x}_k$ must then be augmented by the calibration group membership indicators. The resulting auxiliary data vector will then contain a maximum of $P \times (Q + 1)$ elements. That is,

$$\mathbf{x}_k = \left(I_{1k}, \dots, I_{Pk}, x_{k1}I_{1k}, \dots, x_{k1}I_{Pk}, \dots, x_{kQ}I_{1k}, \dots, x_{kQ}I_{Pk}\right)'. \tag{3.3}$$

Depending on how the calibration was precisely done, not all components of $\mathbf{x}_k$ are required to be present in the vector. For example, if no calibration was done in calibration group p', then the components $I_{p'k}, x_{k1}I_{p'k}, \dots, x_{kQ}I_{p'k}$ corresponding to that calibration group can be omitted from $\mathbf{x}_k$ for each k.

If the sample file is given at the estimation level, the calibration groups would also be at the estimation level and would have to be made consistent with the sampling units. We also define the following notation:

I_{pj} is the calibration group indicator for estimation unit j in calibration group U_p. That is, I_{pj} equals 1 if j is in the domain and 0 otherwise

x_{jq} is the value of the q-th auxiliary variable for estimation unit j, for $j \in k$

To obtain the aggregated auxiliary variable at the sampling level, we compute:

$$\tilde{x}_{kpq} = \sum_{j \in k} I_{pj} x_{jq}. \tag{3.4}$$

To obtain the aggregated count of estimation units in a sampling unit in calibration group p, we calculate:

$$\tilde{I}_{pk} = \sum_{j \in k} I_{pj}. \tag{3.5}$$

It is possible that $\tilde{I}_{pk}$ is greater than 1. The auxiliary data vector at the sampling level can then be written as

$$\mathbf{x}_k = \left(\tilde{I}_{1k}, \dots, \tilde{I}_{Pk}, \tilde{x}_{k11}, \dots, \tilde{x}_{kP1}, \dots, \tilde{x}_{k1Q}, \dots, \tilde{x}_{kPQ}\right)'. \tag{3.6}$$

Finally, we can use Equation (3.6) to obtain the auxiliary data matrix $\mathbf{X}$. Again, if no calibration was done in calibration group p', then the components $\tilde{I}_{p'k}, \tilde{x}_{kp'1}, \dots, \tilde{x}_{kp'Q}$ corresponding to that calibration group can be omitted from $\mathbf{x}_k$ for each k.

Note that (3.6) can be seen as a special case of (3.3) when the estimation and sampling units coincide. In this case, we would have $\tilde{I}_{pj} = I_{pj} = I_{pk}$ and

$$\begin{aligned} \tilde{x}_{kpq} &= I_{pj} x_{jq} \\ &= I_{pk} x_{kq}. \end{aligned}$$

Example 8

Suppose that the CalibrationGroupFile was the following:

CalibrationID	CalibrationGroup	CalibrationGroupExpression	AuxiliaryVariables	…
1	province	revenue > 100000	_N_	
2	naics2		_N_	
3	size		_N_	

The auxiliary data matrix would be formed as follows:

id	province	take_all	province = ‘AB’ and (revenue > 100000)	province = ‘BC’ and (revenue > 100000)	…
1	AB	1	1	0	
2	AB	0	1	0	
3	BC	0	0	1	
4	BC	0	0	1	
5	BC	0	0	0	
6	BC	0	0	0	
7	AB	0	0	0	
8	ON	1	0	0	
9	ON	0	0	0	
10	QC	0	0	0	
…					

Example 9

Now suppose that we also had CalibrationExclusionExpression = *take_all = 1*, the auxiliary data matrix would now be:

id	province	take_all	province = ‘AB’ and (revenue > 100000) and not (take_all = 1)	province = ‘BC’ and (revenue > 100000) and not (take_all = 1)	…
1	AB	1	0	0	
2	AB	0	1	0	
3	BC	0	0	1	
4	BC	0	0	1	
5	BC	0	0	0	
6	BC	0	0	0	
7	AB	0	0	0	
8	ON	1	0	0	
9	ON	0	0	0	
10	QC	0	0	0	
…					

The changed values resulting from including the CalibrationExclusionExpression are highlighted.

In general, the CalibrationGroupExpression in the CalibrationGroupFile is augmented by the negation of the CalibrationExclusionExpression. That is,

```
CalibrationGroupExpression and not (CalibrationExclusionExpression)
```

This would occur for each record in the CalibrationGroupFile. If the sampling design is two-phase, we would have for each CalibrationGroupFile record:

```
CalibrationGroupExpression1 and not (CalibrationExclusionExpression1)
CalibrationGroupExpression2 and not (CalibrationExclusionExpression2)
```

3.4 Linearized Values

Once the extended domain variables have been defined, they can be used in the applicable variance estimation formula. The variables for the total and size can be used directly, but the variables for the mean and ratio must be linearized. Therefore, define u_{dk} according to the estimated domain parameter. We have

Total: $u_{dk} = y_{dk}$

Size: $u_{dk} = I_{dk}$

Mean: $u_{dk} = \left(y_{dk} - \hat{\bar{Y}}_d I_{dk}\right)/\hat{N}_d$, where $\hat{Y}_d = \hat{Y}_d/\hat{N}_d$

Ratio: $u_{dk} = (y_{1dk} - \hat{R}_d y_{2dk})/\hat{Y}_{2d}$, where $\hat{R}_d = \hat{Y}_{1d}/\hat{Y}_{2d}$

The foregoing formulas are applicable in the case where the sampling unit level is the same as the estimation unit level. In the case where the sampling unit level is different from the estimation unit, modifications to these basic formulas are required. The specific modifications depend on whether the count level is at the sampling unit or estimation unit level.

The count level is the estimation unit

Total: $u_{dk} = \sum_{j\in k} y_{dj}$

Size: $u_{dk} = \sum_{j\in k} I_{dj}$

Mean: $u_{dk} = (\sum_{j\in k} y_{dj} - \hat{\bar{Y}}_d \sum_{j\in k} I_{dj})/\sum_{k\in s} w_k \sum_{j\in k} I_{dj}$, where $\hat{Y}_d = \hat{Y}_d/\hat{N}_d$

Ratio: $u_{dk} = (\sum_{j\in k} y_{1dj} - \hat{R}_d \sum_{j\in k} y_{2dj})/\sum_{k\in s} w_k \sum_{j\in k} y_{2dj}$, where $\hat{R}_d = \hat{Y}_{1d}/\hat{Y}_{2d}$

The count level is the sampling unit

Total: $u_{dk} = \sum_{j\in k} y_{dj}$

Size: $u_{dk} = \max_{j\in k} I_{dj}$

Mean: $u_{dk} = (\sum_{j\in k} y_{dj} - \hat{\bar{Y}}_d \max_{j\in k} I_{dj})/\sum_{k\in s} w_k \max_{j\in k} I_{dj}$, where $\hat{Y}_d = \hat{Y}_d/\hat{N}_d$

Ratio: $u_{dk} = (\sum_{j\in k} y_{1dj} - \hat{R}_d \sum_{j\in k} y_{2dj})/\sum_{k\in s} w_k \sum_{j\in k} y_{2dj}$, where $\hat{R}_d = \hat{Y}_{1d}/\hat{Y}_{2d}$

3.5 Residuals

Residuals are computed based on the linearized variable of the parameter to be estimated and the auxiliary data matrix.

3.5.1 *One-Phase Designs*

Given a set of auxiliary variables, the estimated residual $\hat{e}_{dk}$ is computed as

$$\hat{e}_{dk} = u_{dk} - \mathbf{x}'_k \widehat{\mathbf{B}}_{(u:\mathbf{x})}, \tag{3.7}$$

where $\mathbf{x}_k$ is the vector of auxiliary variables for unit k and $\widehat{\mathbf{B}}_{(u:\mathbf{x})}$ is the vector of estimated regression parameters in the linear model relating u_{dk} and $\mathbf{x}_k$. Let c_k be a positive regression adjustment variable that can be specified by the user. To compute $\widehat{\mathbf{B}}_{(u:\mathbf{x})}$, we use the following formula:

$$\widehat{\mathbf{B}}_{(u:\mathbf{x})} = \left(\sum_{k \in s} a_k \mathbf{x}_k \mathbf{x}'_k / c_k \right)^{-1} \sum_{k \in s} a_k \mathbf{x}_k u_{dk} / c_k. \tag{3.8}$$

Since the parameters that we deal with in the first version of `G-Est` are functions of totals, we will see in subsequent sections that the corresponding variance estimators can be written in terms of Equation (3.7).

3.5.2 *Two-Phase Designs*

The two-phase case adds some complexity due to both sampling at two phases as well as calibration at one or both phases. The theoretical basis of the computations that we use is due to Estevao and Särndal (2009). The formulas and notation that we use in this document are also based on that paper.

Depending on which combination of auxiliary variables was used, one or more of the following regression parameters would need to be computed. In this section, the formulas for estimating all of the potential regression parameters are presented. The following section will indicate specifically which regression parameters will be necessary based on the auxiliary information used.

We will call the first phase sample, s_1, and the second phase sample, s. Corresponding to the cases in Table 1.1 of Estevao and Särndal (2009), in `G-Est`, the ability to calibrate to the following combinations of auxiliary information will be available:

1. Totals are known for auxiliary variables available at the first phase (and hence also the second phase). Let these auxiliary variables be denoted by $\mathbf{x}_{1k}$.

2. Totals are known for auxiliary variables available at the second phase only. Let these auxiliary variables be denoted by $\mathbf{x}_{2k}$.

3. Estimated totals using first phase design weights for auxiliary variables available at the first phase. Let these auxiliary variables be denoted by $\mathbf{x}_{k(a)}$.

4. Estimated totals using first phase calibration weights for auxiliary variables available at the first phase. Let these auxiliary variables be denoted by $\mathbf{x}_{k(w)}$.

5. Since totals known for auxiliary variables available at the first phase can also be used at the second phase, we define $\mathbf{x}_{k(t)}$ to be the union of the auxiliary variables $\mathbf{x}_{1k}$ and $\mathbf{x}_{2k}$.

The following development looks at the case of the variance of the estimator of the total $\hat{Y}$. We first define some of the equations that we will need for the final variance formula. Let the first phase design weights be a_{1k} and the final second phase design weights be a_k. That is, if the conditional second phase design weight is given by a_{2k}, then $a_k = a_{1k}a_{2k}$. Then, with the starting design weight a_k, we have the second phase calibration weight, which we write as w_k. Denote by $\widehat{\mathbf{B}}_{(u;\mathbf{x})}$ the estimated regression parameter of the dependent variable y on the auxiliary data vector

$$\mathbf{x}_k = \begin{pmatrix} \mathbf{x}_{k(t)} \\ \mathbf{x}_{k(w)} \\ \mathbf{x}_{k(a)} \end{pmatrix}.$$

The following formula gives the estimated regression parameters based on the auxiliary variables used at the second phase:

$$\widehat{\mathbf{B}}_{(u;\mathbf{x})} = \left(\sum_{k\in s} a_k \mathbf{x}_k \mathbf{x}_k' / c_k\right)^{-1} \sum_{k\in s} a_k \mathbf{x}_k u_{dk} \,/ c_k, \qquad (3.9)$$

where $\mathbf{x}_k = (\mathbf{x}_{k(t)}', \mathbf{x}_{k(w)}', \mathbf{x}_{k(a)}')'$.

Once the vector of estimated regression parameters, $\widehat{\mathbf{B}}_{(u;x)}$, has been computed, it can be partitioned as

$$\widehat{\mathbf{B}}_{(u;x)} = \begin{pmatrix} \widehat{\mathbf{B}}_{(u;\mathbf{x})(t)} \\ \widehat{\mathbf{B}}_{(u;\mathbf{x})(w)} \\ \widehat{\mathbf{B}}_{(u;\mathbf{x})(a)} \end{pmatrix},$$

where each sub-vector corresponds to a set of auxiliary variables in $\mathbf{x}_k$. For example, the estimated regression parameters in $\widehat{\mathbf{B}}_{(u;\mathbf{x})(w)}$ correspond to the auxiliary variables $\mathbf{x}_{k(w)}$. Combining each regression parameter vector with the corresponding auxiliary data vectors, gives us the required predicted values: $\mathbf{x}'_{k(t)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(t)}$, $\mathbf{x}'_{k(w)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(w)}$, and $\mathbf{x}'_{k(a)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(a)}$.

If totals estimated using the phase 1 calibration weights were used, then the following estimated regression parameters of $\mathbf{x}_{k(w)}'\widehat{\mathbf{B}}_{(u;\mathbf{x})(w)}$ based on the auxiliary variables used for phase 1 calibration must also be computed:

$$\widehat{\mathbf{B}}_{(\mathbf{x}\widehat{\mathbf{B}}_w;\mathbf{x}_1)} = \left(\sum_{k\in s_1} a_{1k} \mathbf{x}_{1k} \mathbf{x}_{1k}' / c_{1k}\right)^{-1} \sum_{k\in s_1} a_{1k} \mathbf{x}_{1k} \mathbf{x}_{k(w)}' \widehat{\mathbf{B}}_{(u;\mathbf{x})(w)} \,/ c_{1k}. \qquad (3.10)$$

Using Equations (3.9) and (3.10), we define the following residuals:

$$\hat{e}_{1dk} = \mathbf{x}'_{k(a)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(a)} + \mathbf{x}'_{k(w)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(w)} - \mathbf{x}'_{1k}\widehat{\mathbf{B}}_{(\mathbf{x}\widehat{\mathbf{B}}_w;\mathbf{x}_1)} \tag{3.11}$$

and

$$\begin{aligned}\hat{e}_{2dk} &= u_{dk} - \mathbf{x}'_k\widehat{\mathbf{B}}_{(u;\mathbf{x})} \\ &= u_{dk} - \mathbf{x}'_{k(t)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(t)} - \mathbf{x}'_{k(a)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(a)} - \mathbf{x}'_{k(w)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(w)}.\end{aligned} \tag{3.12}$$

Depending on the auxiliary information actually used in the calibration, not all of the regression parameters will need to be estimated. Consequently, the precise form of $\hat{e}_{1dk}$ and $\hat{e}_{2dk}$ may also change. The following table summarizes the different combinations of auxiliary information that may have been incorporated in calibration weighting and the corresponding form of the residuals, $\hat{e}_{1dk}$ and $\hat{e}_{2dk}$, that is to be computed for the final variance formula.

Table 3.1 Formulas for $\hat{e}_{1dk}$ and $\hat{e}_{2dk}$, based on different combinations of auxiliary information

Auxiliary Variables	$\hat{e}_{1dk}$	$\hat{e}_{2dk}$
$\mathbf{x}_{\boldsymbol{k}(t)}, \mathbf{x}_{\boldsymbol{k}(w)}, \mathbf{x}_{\boldsymbol{k}(a)}$	$\mathbf{x}'_{k(a)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(a)} + \mathbf{x}'_{k(w)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(w)} - \mathbf{x}'_{1k}\widehat{\mathbf{B}}_{(\mathbf{x}\widehat{\mathbf{B}}_w;\mathbf{x}_1)}$	$u_{dk} - \mathbf{x}'_{k(t)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(t)} - \mathbf{x}'_{k(a)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(a)} - \mathbf{x}'_{k(w)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(w)}$
$\mathbf{x}_{k(t)}, \mathbf{x}_{k(w)}$	$\mathbf{x}'_{k(w)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(w)} - \mathbf{x}'_{1k}\widehat{\mathbf{B}}_{(\mathbf{x}\widehat{\mathbf{B}}_w;\mathbf{x}_1)}$	$u_{dk} - \mathbf{x}'_{k(t)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(t)} - \mathbf{x}'_{k(w)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(w)}$
$\mathbf{x}_{k(t)}, \mathbf{x}_{k(a)}$	$\mathbf{x}'_{k(a)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(a)}$	$u_{dk} - \mathbf{x}'_{k(t)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(t)} - \mathbf{x}'_{k(a)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(a)}$
$\mathbf{x}_{k(w)}, \mathbf{x}_{k(a)}$	$\mathbf{x}'_{k(a)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(a)} + \mathbf{x}'_{k(w)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(w)} - \mathbf{x}'_{1k}\widehat{\mathbf{B}}_{(\mathbf{x}\widehat{\mathbf{B}}_w;\mathbf{x}_1)}$	$u_{dk} - \mathbf{x}'_{k(a)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(a)} - \mathbf{x}'_{k(w)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(w)}$
$\mathbf{x}_{k(t)}$	0	$u_{dk} - \mathbf{x}'_{k(t)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(t)}$
$\mathbf{x}_{k(w)}$	$\mathbf{x}'_{k(w)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(w)} - \mathbf{x}'_{1k}\widehat{\mathbf{B}}_{(\mathbf{x}\widehat{\mathbf{B}}_w;\mathbf{x}_1)}$	$u_{dk} - \mathbf{x}'_{k(w)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(w)}$
$\mathbf{x}_{k(a)}$	$\mathbf{x}'_{k(a)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(a)}$	$u_{dk} - \mathbf{x}'_{k(a)}\widehat{\mathbf{B}}_{(u;\mathbf{x})(a)}$

3.6 One-Phase Variance

The variance estimator of the calibration estimator of the domain parameter is then given by

$$\sum_{l\in s}\sum_{k\in s}\frac{a_k a_l - a_{kl}}{a_k a_l}(g_k a_k \hat{e}_{dk})(g_l a_l \hat{e}_{dl}). \tag{3.13}$$

The calibration factor g_k is computed as $g_k = w_k/a_k$. Although this implies that $w_k = g_k a_k$, we will leave the expression $g_k a_k$ in Equation (3.13) for two reasons. The first is that the derivation of (3.13) does not naturally include the term g_k, which is only added afterwards to improve the properties of the estimator. The second is to explicitly indicate that it is possible to specify only the calibration factor rather than the calibration weight.

3.7 Two-Phase Variance

Define $D_{1kl} = a_{1k}a_{1l} - a_{1kl}$ and $D_{kl} = a_k a_l - a_{kl}$. The variance estimator of the two-phase calibration estimator of the domain total, $\sum_{k\in s} w_k y_{dk}$, is given by

$$\begin{aligned}
\hat{V}(\hat{Y}_d) &= \sum_{k\in s_1}\sum_{l\in s_1} D_{1kl}(g_{1k}\hat{e}_{1dk})(g_{1l}\hat{e}_{1dl}) + \sum_{k\in s}\sum_{l\in s} D_{kl}(g_k\hat{e}_{2dk})(g_l\hat{e}_{2dl}) \\
&\quad + 2\sum_{k\in s_1}\sum_{l\in s} D_{1kl}a_{2l}(g_{1k}\hat{e}_{1dk})(g_l\hat{e}_{2dl}) \qquad (3.14) \\
&= v_1 + v_2 + v_3
\end{aligned}$$

where

$$\begin{aligned}
v_1 &= \sum_{k\in s_1}\sum_{l\in s_1} D_{1kl}(g_{1k}\hat{e}_{1dk})(g_{1l}\hat{e}_{1dl}), \\
v_2 &= \sum_{k\in s}\sum_{l\in s} D_{kl}(g_k\hat{e}_{2dk})(g_l\hat{e}_{2dl}), \\
v_3 &= 2\sum_{k\in s_1}\sum_{l\in s} D_{1kl}a_{2l}(g_{1k}\hat{e}_{1dk})(g_l\hat{e}_{2dl}).
\end{aligned}$$

The terms v_1, v_2 and v_3 can be described as the contributions of first phase, of second phase and of the intersection of the two phases, respectively.

Here, as in the one-phase case, we use the weighted residual technique, where we multiply the residuals by the corresponding calibration factor. The calibration factor at phase 1 is $g_{1k} = w_{1k}/a_{1k}$ and the calibration factor at phase 2 is $g_k = w_k/a_k$.

If there was no calibration done at phase 1, we set $g_{1k} = 1$.

The terms of the variance estimator can also be written in terms of the inclusion probabilities:

$$\begin{aligned}
v_1 &= \sum_{k\in s_1}\sum_{l\in s_1} \frac{\pi_{1kl} - \pi_{1k}\pi_{1l}}{\pi_{1kl}} \frac{g_{1k}\hat{e}_{1dk}}{\pi_{1k}} \frac{g_{1l}\hat{e}_{1dl}}{\pi_{1l}} \\
v_2 &= \sum_{k\in s}\sum_{l\in s} \frac{\pi_{1kl}\pi_{2kl} - \pi_{1k}\pi_{2k}\pi_{1l}\pi_{2l}}{\pi_{1kl}\pi_{2kl}} \frac{g_k\hat{e}_{2dk}}{\pi_{1k}\pi_{2k}} \frac{g_l\hat{e}_{2dl}}{\pi_{1l}\pi_{2l}} \qquad (3.15) \\
v_3 &= 2\sum_{k\in s_1}\sum_{l\in s} \frac{\pi_{1kl} - \pi_{1k}\pi_{1l}}{\pi_{1kl}} \frac{g_{1k}\hat{e}_{1dk}}{\pi_{1k}} \frac{g_l\hat{e}_{2dl}}{\pi_{1l}\pi_{2l}}.
\end{aligned}$$

Now one can observe that v_2 can be rewritten as

$$v_2 = \sum_{k\in s}\sum_{l\in s} \frac{\pi_{1kl} - \pi_{1k}\pi_{1l}}{\pi_{1kl}\pi_{2kl}} \frac{g_k\hat{e}_{2dk}}{\pi_{1k}} \frac{g_l\hat{e}_{2dl}}{\pi_{1l}} + \sum_{k\in s}\sum_{l\in s} \frac{\pi_{2kl} - \pi_{2k}\pi_{2l}}{\pi_{2kl}} \frac{g_k\hat{e}_{2dk}}{\pi_{1k}\pi_{2k}} \frac{g_l\hat{e}_{2dl}}{\pi_{1l}\pi_{2l}}. \qquad (3.16)$$

3.8 Special Case of the Horvitz-Thompson Estimator

In either the one-phase or two-phase sampling, if there no calibration was done, the variance formulas can be simplified. In particular, the residuals will not have to be computed. Suppose no calibration is done, that is, the design weights a_k are used in estimation. In one-phase sampling, the variance estimator is given by

$$\hat{V} = \sum_{l \in s} \sum_{k \in s} \frac{a_k a_l - a_{kl}}{a_k a_l} (a_k u_{dk})(a_l u_{dl}). \tag{3.17}$$

In two-phase sampling, from (3.16), the variance estimator is given by

$$\hat{V} = \sum_{k \in s} \sum_{l \in s} \frac{\pi_{1kl} - \pi_{1k}\pi_{1l}}{\pi_{1kl}\pi_{2kl}} \frac{u_{dk}}{\pi_{1k}} \frac{u_{dl}}{\pi_{1l}} + \sum_{k \in s} \sum_{l \in s} \frac{\pi_{2kl} - \pi_{2k}\pi_{2l}}{\pi_{2kl}} \frac{u_{dk}}{\pi_{1k}\pi_{2k}} \frac{u_{dl}}{\pi_{1l}\pi_{2l}}. \tag{3.18}$$

3.9 Variance Formulas for SRSWOR

The previous section gave general formulas for computing the variance of various calibration estimators; however, actual implementation depends on how the calibration weights were derived and which sampling design was used. In this section, formulas are given for computing the variance estimates under simple random sampling without replacement (SRSWOR). More generally, we assume that stratification has taken place, so that computations will be done within stratum.

In SRSWOR, given s_h, the collection of sample units that belong to stratum h, the design weight for element k is given by

$$a_k = N_h / n_h \quad \text{for } k \in s_h.$$

Variance of calibration estimator

Given the values for the population and sample sizes for stratum h, the variance estimator is the following:

$$\Psi(\hat{e}_{dk}) = \sum_{h=1}^{H} N_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{s_h^2}{n_h},$$

where $s_h^2 = \frac{1}{n_h - 1} \sum_{k \in s_h} (g_k \hat{e}_{dk} - \bar{e}_{dh})^2$ and $\bar{e}_{dh} = \frac{1}{n_h} \sum_{k \in s_h} g_k \hat{e}_{dk}$.

Variance of the Horvitz-Thompson estimator

Given the values for the population and sample sizes for stratum h, the variance estimator is the following:

$$\Psi(u_{dk}) = \sum_{h=1}^{H} N_h^2 \left(1 - \frac{n_h}{N_h}\right) \frac{s_h^2}{n_h},$$

where $s_h^2 = \frac{1}{n_h - 1} \sum_{k \in s_h} (u_{dk} - \bar{u}_{dh})^2$ and $\bar{u}_{dh} = \frac{1}{n_h} \sum_{k \in s_h} u_{dk}$.

3.10 Variance Formulas for Poisson

In Poisson sampling, sample units are selected independently of each other and can have a different weight regardless of the stratum. Therefore, the joint probability of selection of units k and l is such that

$$\pi_{kl} = \begin{cases} \pi_k & \text{if } k = l \\ \pi_k \pi_l & \text{if } k \neq l \end{cases}.$$

Variance of calibration estimator

Using Equation (3.13), we obtain the following for the variance estimator of the domain parameter estimate,

$$\Psi(\hat{e}_{dk}) = \sum_{k \in s} (1 - \pi_k) \left(\frac{g_k \hat{e}_{dk}}{\pi_k}\right)^2$$

Rewriting this in terms of design weights, we have

$$\Psi(\hat{e}_{dk}) = \sum_{k \in s} a_k (a_k - 1)(g_k \hat{e}_{dk})^2.$$

Variance of the Horvitz-Thompson estimator

Substituting u_{dk} for $\hat{e}_{dk}$ and setting $g_k = 1$, we obtain

$$\Psi(u_{dk}) = \sum_{k \in s} (1 - \pi_k) \left(\frac{u_{dk}}{\pi_k}\right)^2,$$

or

$$\Psi(u_{dk}) = \sum_{k \in s} a_k (a_k - 1)(u_{dk})^2.$$

3.11 Variance Formulas for SRSWOR-SRSWOR

To obtain the formulas for the variance estimators under two-phase sampling with SRSWOR at each phase, we start with Equations (3.14), (3.15) and (3.16), and substitute the inclusion probabilities under the design. The formula for the two-phase estimator of variance for SRSWOR-SRSWOR described in this section is taken from Generalized Systems Methods (2011).

Let the first phase strata be denoted by h, for $h = 1, \dots, H$, and the second phase strata be denoted by g, for $g = 1, \dots, G$, where the second phase strata g may be different from h and are not necessarily nested within h. Let N_{1h} and n_{1h} denote the population and sample sizes within strata h at phase 1 and N_{2g} and n_{2g} denote the population and sample sizes within strata g at phase 2 strata. Furthermore, let the portion of the phase 1 sample that belongs to stratum h be s_{1h} and the portion of the phase 2 sample that belongs to stratum g be s_g. The inclusion probabilities for the first and second phases are as follows.

The inclusion probability for element k at the phase 1 is given by:

$$\pi_{1k} = \frac{n_{1h}}{N_{1h}} \quad \text{for } k \in s_{1h}. \tag{3.19}$$

The joint inclusion probability of elements k and l at phase 1 is given by:

$$\pi_{1kl} = \begin{cases} \dfrac{n_{1h}}{N_{1h}} & \text{for } k = l \text{ and } k \in s_{1h} \\ \dfrac{n_{1h}}{N_{1h}} \dfrac{n_{1h} - 1}{N_{1h} - 1} & \text{for } k \neq l \text{ and } k \in s_{1h}, l \in s_{1h} \\ \dfrac{n_{1h}}{N_{1h}} \dfrac{n_{1h'}}{N_{1h'}} & \text{for } k \neq l \text{ and } k \in s_{1h}, l \in s_{1h'}, h \neq h' \end{cases}. \tag{3.20}$$

The inclusion probability of element k at phase 2 is given by:

$$\pi_{2k} = \frac{n_{2g}}{N_{2g}} \quad \text{for } k \in s_g \text{ and } k \in s_1. \tag{3.21}$$

The joint inclusion probabilities of elements k and l are given by:

$$\pi_{2kl} = \begin{cases} \dfrac{n_{2g}}{N_{2g}} & \text{for } k = l \text{ and } k \in s_g \\ \dfrac{n_{2g}}{N_{2g}} \dfrac{n_{2g} - 1}{N_{2g} - 1} & \text{for } k \neq l \text{ and } k \in s_g, l \in s_g \\ \dfrac{n_{2g}}{N_{2g}} \dfrac{n_{2g'}}{N_{2g'}} & \text{for } k \neq l \text{ and } k \in s_g, l \in s_{g'}, g \neq g' \end{cases}. \tag{3.22}$$

In addition, let $f_{1h} = n_{1h}/N_{1h}$ denote the sampling fraction in the first phase stratum and $f_{2g} = n_{2g}/N_{2g}$ denote the sampling fraction in the second phase stratum, given inclusion in the first phase sample, s_1.

Variance of the calibration estimator

Using the inclusion probabilities above, the three terms of the variance v_1, v_2 and v_3 from section 3.7 become after tedious computations:

$$v_1 = \sum_{h=1}^{H} \frac{N_{1h}^2}{n_{1h}} (1 - f_{1h}) s_{e_1,h}^2, \tag{3.23}$$

where $s_{e_1,h}^2 = \frac{1}{n_{1h}-1} \sum_{k \in s_{1h}} (g_{1k}\hat{e}_{1dk} - \bar{e}_{1h})^2$ and $\bar{e}_{1h} = \frac{1}{n_{1h}} \sum_{k \in s_{1h}} g_{1k}\hat{e}_{1dk}$.

The term v_1 is simply the variance of the total under SRSWOR. For a given first-phase stratum h, the term $s_{e_1}^2$ is the sample variance of the $\hat{e}_{1dk}$, computed over all elements in s_{1h}.

$$\begin{aligned} v_2 &= \sum_{h=1}^{H} \frac{N_{1h}^2}{n_{1h}} \frac{1 - f_{1h}}{n_{1h} - 1} \left\{ \sum_{g=1}^{G} \sum_{k \in s_{hg}} \frac{N_{2g}}{n_{2g}} g_k^2 \hat{e}_{2dk}^2 - \frac{1}{n_{1h}} \left[\sum_{g=1}^{G} \sum_{k \in s_{hg}} \frac{N_{2g}}{n_{2g}} g_k \hat{e}_{2dk} \right]^2 \right\} \\ &+ \sum_{h=1}^{H} \frac{N_{1h}^2}{n_{1h}^2} \frac{1 - f_{1h}}{n_{1h} - 1} \left\{ \sum_{g=1}^{G} \frac{N_{2g}^2}{n_{2g}} (1 - f_{2g}) s_{\delta_2}^2 \right\} + \sum_{g=1}^{G} \frac{N_{2g}^2}{n_{2g}} (1 - f_{2g}) s_{\zeta_2}^2 \\ &= v_{21} + v_{22} + v_{23} + v_{24}, \end{aligned} \tag{3.24}$$

where

$$v_{21} = \sum_{h=1}^{H} \frac{N_{1h}^2}{n_{1h}} \frac{1 - f_{1h}}{n_{1h} - 1} \left\{ \sum_{g=1}^{G} \sum_{k \in s_{hg}} \frac{N_{2g}}{n_{2g}} g_k^2 \hat{e}_{2dk}^2 \right\},$$

$$v_{22} = -\sum_{h=1}^{H} \frac{N_{1h}^2}{n_{1h}} \frac{1 - f_{1h}}{n_{1h} - 1} \frac{1}{n_{1h}} \left\{ \left[\sum_{g=1}^{G} \sum_{k \in s_{hg}} \frac{N_{2g}}{n_{2g}} g_k \hat{e}_{2dk} \right]^2 \right\},$$

$$v_{23} = \sum_{h=1}^{H} \frac{N_{1h}^2}{n_{1h}^2} \frac{1 - f_{1h}}{n_{1h} - 1} \left\{ \sum_{g=1}^{G} \frac{N_{2g}^2}{n_{2g}} (1 - f_{2g}) s_{\delta_2}^2 \right\},$$

$$v_{24} = \sum_{g=1}^{G} \frac{N_{2g}^2}{n_{2g}} (1 - f_{2g}) s_{\zeta_2}^2,$$

and $s_{\delta_2}^2 = \frac{1}{n_{2g}-1} \sum_{k \in s_g} \left(\hat{\delta}_{2dk} - \bar{\delta}_{2g}\right)^2$, $\bar{\delta}_{2g} = \frac{1}{n_{2g}} \sum_{k \in s_g} \hat{\delta}_{2dk}$, and $\hat{\delta}_{2dk} = \begin{cases} g_k \hat{e}_{2dk} & \text{if } k \in s_{1h} \\ 0 & \text{otherwise} \end{cases}$,
and $s_{\zeta_2}^2 = \frac{1}{n_{2g}-1} \sum_{k \in s_g} \left(\hat{\zeta}_{2dk} - \bar{\zeta}_{2g}\right)^2$, $\bar{\zeta}_{2g} = \frac{1}{n_{2g}} \sum_{k \in s_g} \hat{\zeta}_{2dk}$, and $\hat{\zeta}_{2dk} = \frac{N_{1h}}{n_{1h}} g_k \hat{e}_{2dk}$ for $k \in s_{1h}$.

In v_2, for the first three subcomponents, there are two summuations over the first and second phase strata, h and g, respectively. For a given first phase stratum h and second phase stratum g, the set of second phase elements in both h and g is denoted by s_{hg}.

In v_{23}, we can see that the variance estimate of the total of $\hat{\delta}_{2dk}$ under SRSWOR is embedded in the calculation.

The term ν_{24} is the variance estimate under SRSWOR of the variable $\hat{\zeta}_{2dk}$. Although $g_{2k}\hat{e}_{2dk}$ is multiplied by the first phase design weight in stratum h, $\hat{\zeta}_{2dk}$ does not need to be set to zero for elements not selected in the second phase sample since we only require the n_{2g} terms for each g.

$$\nu_3 = 2\sum_{h=1}^{H}\frac{N_{1h}^2}{n_{1h}}(1-f_{1h})\left\{\frac{1}{n_{1h}-1}\sum_{k\in s_{1h}}(g_{1k}\hat{e}_{1dk}-\bar{e}_{1h})(\hat{\epsilon}_{2dk}-\bar{\epsilon}_{2h})\right\}, \tag{3.25}$$

where $\bar{\epsilon}_{2h} = \frac{1}{n_{1h}}\sum_{k\in s_{1h}}\hat{\epsilon}_{2dk}$ and $\hat{\epsilon}_{2dk} = \begin{cases}\frac{N_{2g}}{n_{2g}}g_k\hat{e}_{2dk} & \text{if } k\in s_1 \text{ and } k\in s_g \\ 0 & \text{if } k\in s_1 \text{ and } k\notin s\end{cases}$.

The term ν_3 is comprised of the sample covariance of $g_{1k}\hat{e}_{1dk}$ and $\hat{\epsilon}_{2dk}$, with a multiplicative factor and summed over all the first-phase strata. The quantity $\hat{\epsilon}_{2dk}$ is needed because $(N_{2g}/n_{2g})g_{2k}\hat{e}_{2dk}$ is only available in the second-phase sample. After determining $(N_{2g}/n_{2g})g_{2k}\hat{e}_{2dk}$ for each g, if an element was selected at the first phase, but not the second phase, a zero is assigned to the element.

The variance estimate of a given parameter estimate is then given by combining the three terms above:

$$\Psi(\hat{e}_{1dk},\hat{e}_{2dk}) = \nu_1+\nu_2+\nu_3.$$

Variance of the Horvitz-Thompson estimator

With the Horvitz-Thompson estimator, there is no residual of first-phase (it is equal to zero everywhere), therefore the terms ν_1 and ν_3 become equal to zero. What remains is the term ν_2, where $g_k\equiv 1$ and the residuals $\hat{e}_{2k}$ are replaced with u_{dk}. We have

$$\begin{aligned}\nu_2 = &\sum_{h=1}^{H}\frac{N_{1h}^2}{n_{1h}}\frac{1-f_{1h}}{n_{1h}-1}\left\{\sum_{g=1}^{G}\sum_{k\in s_{hg}}\frac{N_{2g}}{n_{2g}}u_{dk}^2-\frac{1}{n_{1h}}\left[\sum_{g=1}^{G}\sum_{k\in s_{hg}}\frac{N_{2g}}{n_{2g}}u_{dk}\right]^2\right\}\\ &+\sum_{h=1}^{H}\frac{N_{1h}^2}{n_{1h}^2}\frac{1-f_{1h}}{n_{1h}-1}\left\{\sum_{g=1}^{G}\frac{N_{2g}^2}{n_{2g}}(1-f_{2g})s_{\delta_2}^2\right\}+\sum_{g=1}^{G}\frac{N_{2g}^2}{n_{2g}}(1-f_{2g})s_{\zeta_2}^2,\end{aligned} \tag{3.26}$$

where $s_{\delta_2}^2 = \frac{1}{n_{2g}-1}\sum_{k\in s_g}\left(\hat{\delta}_{2dk}-\bar{\delta}_{2g}\right)^2$, $\bar{\delta}_{2g} = \frac{1}{n_{2g}}\sum_{k\in s_g}\hat{\delta}_{2dk}$, and $\hat{\delta}_{2dk} = \begin{cases}u_{dk} & \text{if } k\in s_{1h}\\ 0 & \text{otherwise}\end{cases}$.

and where $s_{\zeta_2}^2 = \frac{1}{n_{2g}-1}\sum_{k\in s_g}\left(\hat{\zeta}_{2dk}-\bar{\zeta}_{2g}\right)^2$, $\bar{\zeta}_{2g} = \frac{1}{n_{2g}}\sum_{k\in s_g}\hat{\zeta}_{2dk}$, and $\hat{\zeta}_{2dk} = \frac{N_{1h}}{n_{1h}}u_{dk}$ for $k\in s_{1h}$.

The two-phase variance of the Horvitz-Thompson estimator is then given by

$$\Psi(u_{dk}) = \nu_2.$$

3.12 Variance Formulas for Poisson-Poisson

To obtain the formulas for the variance estimators under two-phase sampling with Poisson sampling at each phase, we start with Equations (3.15) and substitute the inclusion probabilities under the design. The inclusion probabilities for the first and second phases are as follows.

The inclusion probability of element k at the first phase is given by π_{1k}. The joint probability of k and l at the first phase is given by

$$\pi_{1kl} = \begin{cases} \pi_{1k} & \text{if } k = l \\ \pi_{1k}\pi_{1l} & \text{if } k \neq l \end{cases}. \tag{3.27}$$

The inclusion probability of element k at the second phase, given the first phase sample, is given by $\pi_{k|s_1}$. The joint probability of k and l at the second phase, given the first phase sample, is given by

$$\pi_{2kl} = \begin{cases} \pi_{2k} & \text{if } k = l \text{ and } k \in s_1 \\ \pi_{2k}\pi_{2l} & \text{if } k \neq l \text{ and } k, l \in s_1 \end{cases}. \tag{3.28}$$

Given the foregoing inclusion probabilities, for $k \neq l$, all summands of the three terms of equations (3.15) are equal to zero. Hence, we only require using the inclusion probabilities where $k = l$.

Variance of the calibration estimator

We have the following for the variance estimator under two-phase sampling with Poisson sampling at each phase.

$$\begin{aligned} \Psi(\hat{e}_{1dk}, \hat{e}_{2dk}) = & \sum_{k \in s_1} (1 - \pi_{1k}) \left(\frac{g_{1k}\hat{e}_{1dk}}{\pi_{1k}} \right)^2 + \sum_{k \in s} (1 - \pi_{1k}\pi_{2k}) \left(\frac{g_k \hat{e}_{2dk}}{\pi_{1k}\pi_{2k}} \right)^2 \\ & + 2 \sum_{k \in s} (1 - \pi_{1k}) \left(\frac{g_{1k}\hat{e}_{1dk}}{\pi_{1k}} \right) \left(\frac{g_k \hat{e}_{2dk}}{\pi_{1k}\pi_{2k}} \right). \end{aligned} \tag{3.29}$$

We can also write this expression in terms of the design weights:

$$\begin{aligned} \Psi(\hat{e}_{1dk}, \hat{e}_{2dk}) = & \sum_{k \in s_1} a_{1k}(a_{1k} - 1)(g_{1k}\hat{e}_{1dk})^2 + \sum_{k \in s} a_{1k}a_{2k}(a_{1k}a_{2k} - 1)(g_k \hat{e}_{2dk})^2 \\ & + 2 \sum_{k \in s} a_{1k}a_{2k}(a_{1k} - 1)(g_{1k}\hat{e}_{1dk})(g_k \hat{e}_{2dk}) \\ = & \nu_1 + \nu_2 + \nu_3 \end{aligned} \tag{3.30}$$

where

$$\nu_1 = \sum_{k \in s_1} a_{1k}(a_{1k} - 1)(g_{1k}\hat{e}_{1dk})^2 \ , \tag{3.31}$$

$$\nu_2 = \sum_{k \in s} a_{1k} a_{2k} (a_{1k} a_{2k} - 1)(g_k \hat{e}_{2dk})^2,$$
$$\nu_3 = 2 \sum_{k \in s} a_{1k} a_{2k} (a_{1k} - 1)(g_{1k} \hat{e}_{1dk})(g_k \hat{e}_{2dk}).$$

Variance of the Horvitz-Thompson estimator

Setting $\hat{e}_{1dk} = 0$ and $g_k = 1$ and replacing $\hat{e}_{2dk}$ by u_{dk}, we obtain $\nu_1 = \nu_3 = 0$ and

$$\Psi(u_{dk}) = \sum_{k \in s} (1 - \pi_{1k} \pi_{2k}) \left(\frac{u_{dk}}{\pi_{1k} \pi_{2k}} \right)^2 = \nu_3 \tag{3.32}$$

or, in terms of first and second phase design weights:

$$\Psi(u_{dk}) = \sum_{k \in s} a_{1k} a_{2k} (a_{1k} a_{2k} - 1) u_{dk}^2 \,. \tag{3.33}$$

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